

☎ (206) 296-5864
✉ mendible@seattleu.edu
📱 mendible
🆔 0000-0001-7820-3658

Ariana Mendible, Ph.D.
ASSISTANT PROFESSOR OF MATHEMATICS
SEATTLE UNIVERSITY
901 12th Ave, Seattle WA 98122

Research interests: Social justice applications of data science, reduced-order modeling, data-driven modeling, machine learning, modal decompositions, transport-dominated physics, traveling waves, fluid dynamics

Education

University of Washington

2016-2021

Ph.D. in Mechanical Engineering, M.S. 2018

Seattle, WA

Thesis: Data-Driven Reduced-Order Modeling for Symmetrical Systems

Advisors: Steven Brunton (Mechanical Engineering) and J. Nathan Kutz (Applied Mathematics)

Seattle University

2012-2016

B.S. in Mechanical Engineering, Minor in Mathematics

Seattle, WA

Senior Design project with Ingersoll Rand, resulting in patent

Research advised by John Carter (Mathematics), supported by Clare Boothe Luce Foundation Undergraduate Research Award

Teaching

Instructor

DATA 5310	Data Visualization	F23, F22
DATA 5901	Capstone I	W26 ² , W25, W24, W23
DATA 5902	Capstone II	S26, S25, S24, S23
DATA 5322	Statistical Machine Learning II	S26*, S25 ² , S24, S23, S22
DATA 3310	Data Visualization	W26, W24, F22
MATH 1335	Calculus II	W22 ² , F21 ²

Bold face indicates a new course development, * indicates online modality, ² indicates two concurrent sections

Capstone Advisor

The Climate Pledge Recruitment Propensity	MSDS 26.1
Amazon Worldwide Sustainability	
Planning and Mission Effectiveness Demographic Data History	MSDS 25.2
Archdiocese of Seattle	
A Modern Approach to Pacific Salmon Research: An Online Dashboard	MSDS 25.5
NOAA Fisheries	
Meta-Analysis of Elwha River Water Temperature Regime	MSDS 24.3
NOAA Fisheries, Fish Ecology Division	
Washington State Craft Brewing Industry Snapshot	MSDS 24.5
Washington Brewers Guild	
Donor Database Analysis and Recommendations	MSDS 23.1
Camp Korey	
A Data Curation System and Predictive Model for Unconfined Concrete Strength	MSDS 23.3
DeSimone Consulting Engineers	

Publications

In Progress

Community-Driven Data Science Practices A Barreda II, C Díaz Eaton, S Hansen, JE Hibdon, L Gordon, R Greenwald, M José Gutiérrez Paz, K Ínce, C Kelling, A Mendible , J Mercado, Victor Piercey, Bianca Thompson Preprint DOI: 10.48550/arXiv.2602.00356	<i>La Matematica</i> Accepted 2026
The Geography of Nobel Prize Selection CM Topaz, A Kurongonayini, B Ptak, A Fluehr, A Mendible , R Roca, N Rodríguez, L Xian, R Schwartz, J Higdon Preprint DOI: 10.1080/2330443X.2024.2379270	<i>Quant Sci Studies</i> Under review 2026
Evaluating Remote Sensing and Machine Learning Methods for Predicting Understory Biomass in Post-Harvest Forests SE Dixon, RJ McGaughey, A Mendible , BT Bormann, C Bobsin and GJ Ettl Preprint DOI: 10.20944/preprints202607.0815.v1	<i>Remote Sensing</i> Under review 2026

Peer-Reviewed Journal Publications

Data collection and analysis for small-town policing: challenges and recommendations C Kelling, A Haensch, A Mendible , S Brooks, A Wiedemann, M Aminian, W Hasty, J Higdon DOI: 10.1080/2330443X.2024.2379270	<i>Statistics and Public Policy</i> 2024
Data science and social justice in the mathematics community Q Jones, AR Vindas Meléndes, A Mendible , M Aminian, HZ Brooks, N Alexander, C Díaz Eaton, P Chodrow DOI:10.1090/noti2773	<i>Notices of the Am Math Soc</i> 2023
Data-driven modeling of two-dimensional detonation wave fronts A Mendible , W Lowrie, SL Brunton, JN Kutz DOI: 10.1016/j.wavemoti.2021.102879	<i>Wave Motion</i> 2022
Data-driven modeling of rotating detonation waves A Mendible , J Koch, H Lange, SL Brunton, JN Kutz DOI: 10.1103/PhysRevFluids.6.050507	<i>Phys Rev Fluids</i> 2021
Dimensionality reduction and reduced-order modeling for traveling wave physics A Mendible , SL Brunton, AY Aravkin, W Lowrie, JN Kutz DOI: 10.1007/s00162-020-00529-9	<i>Theor Comput Fluid Dyn</i> 2020
Reduced strain mechanochemical activation onset in microstructured materials JJ Schwartz, R Behrou, B Cao, M Bassford, A Mendible , C Shaeffer, AJ Boydston, N Boechler DOI: 10.1039/C9PY01875A	<i>Polym Chem</i> 2020
Randomized nonnegative matrix factorization NB Erichson, A Mendible , S Wihlbörn, JN Kutz DOI: 10.1016/j.patrec.2018.01.007	<i>Pattern Recognit Lett</i> 2018

Professional & Community-Facing Articles

Technical Appendix to Connecting the Dots: A Preliminary Network Analysis of Complaints about Police in Minneapolis M Wang, A Mendible , C Kelling	<i>DSPACE Blog</i> 2026
Connecting the Dots: A Preliminary Network Analysis of Complaints about Police in Minneapolis C Kelling, A Mendible , M Wang, A Wiedemann, T Marrinan, E Pederson	<i>DSPACE Blog</i> 2025
Counting on Justice: Bridging Quantitative Research and Community Empowerment A Mendible , C Kelling	<i>MAA Focus</i> 2024
Quantitative Justice: Using Data Science for Good A Mendible	<i>MAA Math Values</i> 2024
Getting Away From One Right Answer A Mendible	<i>COMAP Consortium Newsletter</i> 2022

Small Town Police Accountability Lab
The Geography of Nobel Prize Selection

Github
Github

Patents

Air Flow Hour Meter

US Patent 10,713,858

A Mendible, S Heard, T Tran, A Hardesty

2017

Presentations

Quantitative Justice Through R: Data Provenance

AMS Engaged Pedagogy Series

Upcoming 2026

Virtual

Data Science Tools for Community-Driven Police Accountability

NAM-SIAM-AMS Special Session on Quantitative Justice, Joint Mathematical Meetings

Jan 2025

Seattle, WA

Data Science Tools for Community-Driven Police Accountability

Quantitative Justice through Applied Data Science Minisymposium, SIAM Annual Meeting

Jul 2024

Spokane, WA

Panel: Decoding Ethics: Perspectives on Responsible Data Science

Women In Data Science Puget Sound

May 2024

Seattle, WA

Small Town Police Accountability: Case Studies and Opportunities in Data Science

NAM-SIAM-AMS Special Session on Quantitative Justice, Joint Mathematical Meetings

Jan 2024

San Francisco, CA

Data Science for Justice: Small Town Policing

QSIDE Datathon4Justice, Opening Plenary

Nov 2023

Virtual

Small Town Police Accountability: A Data Science Toolkit

SciPy: Scientific Computing with Python

Jul 2023

Austin, TX

Using Data Science for Social Justice

Annual Meeting of the PNW Section of the Mathematical Association of America

Mar 2023

Newberg, OR

QSIDE's STOPA (Small Town Policing Accountability) Research Lab

QSIDE Datathon4Justice, Opening Plenary

Oct 2022

Virtual

Meet Your Data Science Faculty

Seattle University Mathematics Seminar

Oct 2022

Seattle, WA

The Role of QSIDE's STOPA Research Lab in the Field of Criminal Justice

QSIDE Data4Justice Conference

Apr 2022

Virtual

Reduced-Order Modeling

Pacific Lutheran University Mathematics Seminar

Mar 2022

Virtual

Data-driven Modeling of Shock Waves in Rotating Detonation Engines

SIAM Computational Science and Engineering

Mar 2021

Virtual

Data-driven Modeling of Traveling Waves

University of Delaware SIAM Seminar Series

Dec 2020

Virtual

Data-driven Modeling of Traveling Waves

Steve Brunton channel video abstract (14k views)

2020

YouTube

Data-driven Modeling of Detonation Wave Interactions in Rotating Detonation Engines

73rd Annual Meeting of the APS Division of Fluid Dynamics

Nov 2020

Virtual

Dimensionality Reduction and Reduced Order Modeling for Traveling Wave Physics

72nd Annual Meeting of the APS Division of Fluid Dynamics

Nov 2019

Seattle, WA

Poster: The Space-Time Problem for Model Reduction of Traveling Waves

DTRA Technical Program Review

Jun 2019

Fairfax, VA

The Space-time Problem with Model Reduction for Traveling Waves

SIAM Computational Science and Engineering

Mar 2019

Spokane, WA

Scholarly Development

Grant Funding

Comparative Police Complaint Network Analysis

\$7,000

University of Minnesota Institute for Advanced Study

2026

Research and Creative Collaborative with Reinvestigation Workgroup, Co-PI

Optimization of Cryoprotective Agents for Single-Mode Electromagnetic Resonant (SMER)

\$50,000

Rewarming Technology to Enable Long-Term Organ Preservation

Seattle U College of Science and Engineering

2023

Multidisciplinary Research Program Award with Shen Ren (Mechanical Engineering)

Small Town Police Accountability Lab

Workshop hosting

Banff International Research Station for Mathematical Innovation and Discovery

2022

Proposal accepted

Pending and Attempted Funding Applications

Fostering Opportunity, Readiness and Growth for Entry (FORGE) Into STEM Careers

\$2,000,000

National Science Foundation

Under review 2026

Scholarships in Science, Technology, Engineering, and Mathematics (S-STEM) Program, Co-PI

Comparative Police Complaint Network Analysis: Supporting Undergraduate Researchers in Community-Engaged Data Science

\$1,000

American Statistical Association

Under review 2026

Bridging Data & Community Micro Grant

Development of Integrated Data Science Tools Towards Understanding of Perceptions of Policing Through Public Data

\$364,054

National Institute of Justice

2024

Measurement of Community Perceptions, PI

Workshops

Reproducible Machine Learning Workflows for Scientists

Aug 2025

US Research Software Sustainability Institute, Funded Participant

Madison, WI

Community Engaged Learning Faculty Fellowship Workshop

Sep 2023

Seattle U Center for Community Engagement, Funded Participant

Seattle, WA

Educating at the Intersection of Data Science and Social Justice Workshop

Jul 2023

Institute for Computational and Experimental Research in Mathematics, Brown University, Funded Participant

Providence, RI

Community Engaged Learning Workshop Series

May 2023

Seattle U Center for Community Engagement

Seattle, WA

Data Science and Social Justice: Networks, Policy, and Education Workshop

Jun-Jul 2022

Institute for Computational and Experimental Research in Mathematics, Brown University, Funded Participant

Providence, RI

Project NExT Fellow

2021-2022

Mathematical Association of America

Virtual

Preparing for Careers in Teaching Statistics and Data Science Workshop

Aug 2021

Section on Statistics and Data Science Education of the American Statistical Association

Virtual

EPSRC Summer School: Modal Decompositions in Fluid Mechanics

Aug 2019

Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Funded Participant

Cambridge, UK

Research Students

Jesse Loi

AY25-26

Data Accessibility for Public Accountability

MS Data Science

supported by SU Technology Ethics Initiative Student-Scholar Program, poster presentations at SRRCON 2026 and SciPy 2026

David Stanko and Mark Daza

FQ25

Applications of Language Models for Complaint Data: NER and LLMs

MS Data Science

JULY 2026

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4

Emma Oriol	AY23-24
Historical Redlining and Modern Gentrification	MS Data Science
Alexander Churhill	AY23-24
Historical Redlining and Housing Quality in Seattle	MS Data Science
Chang He	AY23-24
Historical Redlining and Cycling Accessibility	MS Data Science
Malia Cortez	AY23-24
Historical Redlining and Population Flows	MS Data Science

Service

Academic Service

Semester Transition Coordinator, Data Science	2025-present
Lead transition from quarters to semesters for data science programs	
MS Data Science Online Course Creation	2025-present
Create material for four asynchronous courses for online MSDS program	
Chile Study Abroad Trip Co-Lead	2024, 2026
Assisted in supervising spring break student trip for Mathematical Models of Near-Shore Phenomena course	
Math Department Recruitment, Inclusion, Diversity, Equity and Retention Committee	2021-present
Member, led climate survey creation and distribution	
QSIDE Institute Consortium SU Liaison	2021-present
Lead event coordination, advertisement at SU, and student involvement	
Bannan Building Safety Committee	2025-present
Math Department Community Building	
Organize end-of-year student celebration for graduating MSDS cohort	2023, 2024, 2025, 2026
Organize weekly (AY 22-23) or quarterly (AY 23-24) activities and events to build departmental community	2022-2024
Math Department Website Review	2025
MS Data Science Student Admissions Review	2022-2025
Review ~250 student applications per year, recommended admissions decisions	
Data Science Faculty Hiring Committee	
Hosted 6 all-day candidates, 2 hires	2025
Reviewed 54 candidate applications, 14 screening interviews, hosted 4 all-day candidates, 2 hires	2023
Reviewed ~9 candidate applications, 9 screening interviews, hosted 2 all-day candidates, position unfilled	2022

Professional Service

QSIDE Data Science, Police Accountability and Community Engagement Research Lab	2021- present
Co-Director, lead meetings and organize research activities	
SciPy Conference	
Program Committe, organize topics and oversee tracks conference-wide	2025, 2026
Track Chair: Human Networks, Social Sciences, and Economics	2024
DSPACE Datathon	2026
Organizer, led two hybrid datathons held between three campuses	
Women In Data Science Puget Sound	
Invited mentor for Mentorship Night for career-seeking women	2024
QSIDE Datathon4Justice	
Co-Organizer	2023
Team Lead	2021, 2022, 2023
Journal Referee	
Physical Review Fluids	
La Matematica	
Humanities and Social Sciences Communications	
Journal of Fluid Mechanics	

Community Service

Mechanical Engineering Students Against Racism , UW Mechanical Engineering	2020-2021
Crafters Against CoVID-19	2020
Engineering Discovery Days , UW College of Engineering	2016, 2019
Bike & Build Drift West	2017
UW Society of Women Engineers , Elementary school outreach visits	2016

Honors & Awards

Community Engaged Learning Faculty Fellowship , Seattle U Center for Community Engagement	AY 23-24
Supervisor of the Year , Seattle University Student Employment Shine Award	2023
Project NExT Fellow , Mathematical Association of America	2021
Graduate Diversity Fellowship , UW Graduate Opportunities and Minority Achievement Program, Full Support	2016, 2020
Steve and Lynn Pratt Fellowship , UW Mechanical Engineering	2017, 2020
Wynne Alexander Guy Spirit of the Mathematics Department Award , SU Mathematics	2019
Wei Li and Dongmei Chen Endowed Fellowship , UW Mechanical Engineering	2017
Brian and Mary Fabien Endowed Fellowship , UW Mechanical Engineering	2016, 2017
Undergraduate Research Award , Clare Boothe Luce Foundation	2014

Additional Experience

Clear Motion (formerly Levant Power)	June - Aug 2016
Controls Engineering Intern	Woburn, MA
National Renewable Energy Laboratory	June - Aug 2015
Science Undergraduate Laboratory Internship	Boulder, CO
Network for Earthquake Engineering Simulation	June - Aug 2014
OH Hinsdale Wave Research Laboratory REU	Corvallis, OR